

SPHCOFEM: Solver for coupling SPH and FE

Luděk Hynčík
New Technologies – Research Centre, University of West Bohemia
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Motivation

Modelling of fluid flow is extremely important in many technical fields including energetics or automotive, but also in many fields of public interest such as biomechanics, bioengineering and clinical applications. Virtual prototyping approach is a standard way to design final products.

Method

The base of the smoothed particle hydrodynamics (SPH) is interpolation theory using kernel defined as

$$\lim_{h \rightarrow 0} w(\mathbf{r}, h) = \delta(\mathbf{r}).$$

The kernel estimate of physical field value $A(\mathbf{r})$ is defined by

$$\langle A(\mathbf{r}) \rangle = \int A(\mathbf{r}') w(\mathbf{r} - \mathbf{r}', h) d\mathbf{r}' \cong \sum_{j=1}^N \frac{m_j}{\rho_j(\mathbf{r}_j)} A(\mathbf{r}_j) w(\mathbf{r} - \mathbf{r}_j, h)$$

considering values of physical field $A(\mathbf{r})$ known on a set of N discrete points given by position vectors and masses. The equations of motion (EOM) together with the equation of state (EOS) form the system to be solved:

$$\frac{d\mathbf{v}_i}{dt} = - \sum_{j=1}^N m_j \left(\frac{p_j}{\rho_j^2} + \frac{p_i}{\rho_i^2} + \Pi_{ij} \right) \nabla w_{ij}, \quad \Delta p = f(\rho, \rho_0, u, \gamma, \kappa)$$

$$\frac{d\rho_i}{dt} = - \sum_{j=1}^N m_j (\mathbf{v}_i - \mathbf{v}_j) \nabla w_{ij}, \quad \frac{du_i}{dt} = - \sum_{j=1}^N m_j \frac{p_j}{\rho_j^2} (\mathbf{v}_i - \mathbf{v}_j) \nabla w_{ij}$$

SPH depends on fast techniques for finding the neighboring particles that contributes to the summations. Nearest neighbor search (NNS) is also the main issue related to the complexity and time use of the SPH calculation and concerns:

- To assess all pairs of particles (N^2).
- To compose particles into sub-sets (subs).

The Lagrange's equations are used to develop EOM for the boundary elements based on the finite element method (FEM).

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\mathbf{q}}_i} \right) - \frac{\partial L}{\partial \mathbf{q}_i} = \mathbf{Q} \quad \mathbf{M}\ddot{\mathbf{q}} + \mathbf{B}\dot{\mathbf{q}} + \mathbf{K}\mathbf{q} = \mathbf{Q}$$

The contact between SPH and FE concerns penalty method and automatic force calculation. The stable time step scheme taking into account the Courant-Friedrichs-Lewy conditions for SPH and stability condition for FEM is applied:

$$\Delta t \leq \min_{FE, SPH} \left\{ k_{stab} \min_{FE} \frac{L_k}{c_k}, \frac{h_j}{c_j + 0.6(\alpha c_j + \beta \mu_j)} \right\}$$

X-SPH with the constant ε between 0 and 1 improves the quality of results:

$$\frac{d\mathbf{x}_i}{dt} = \hat{\mathbf{v}}_i = \mathbf{v}_i + \varepsilon \sum_{j=1}^N m_j \frac{\mathbf{v}_{ij}}{\rho_{ij}} w_{ij}$$

Both EOM for SPH and FE are implemented using C++ and MATLAB to the single SPHCOFEM algorithm:

- SPH: the central acceleration method
- FE: conjugate gradient method

$$F_n = \frac{m_p m_e}{m_p + m_e} \frac{p}{\Delta t^2}$$

The code and the validation

There is a structured input text file describing the task to be investigated by the code. Due to the pre- and post-processor developed as GUI in MATLAB, the M-file can be used to design the structure is also defined.

Shock tube

- Stationary gas divided by diaphragm into two regions
- Initial pressures and densities

Broken dam

- Free surface flow
- Sensitivity analysis

Crashing ball

- Full 3D problem

Results

SPHCOFEM investigating the problems of fluid flow and its interaction with deformable boundaries was developed.

The core of the package in C++ is the Lagrangean method for fluid flow simulation by SPH that is coupled to boundary FE via continuous contact model.

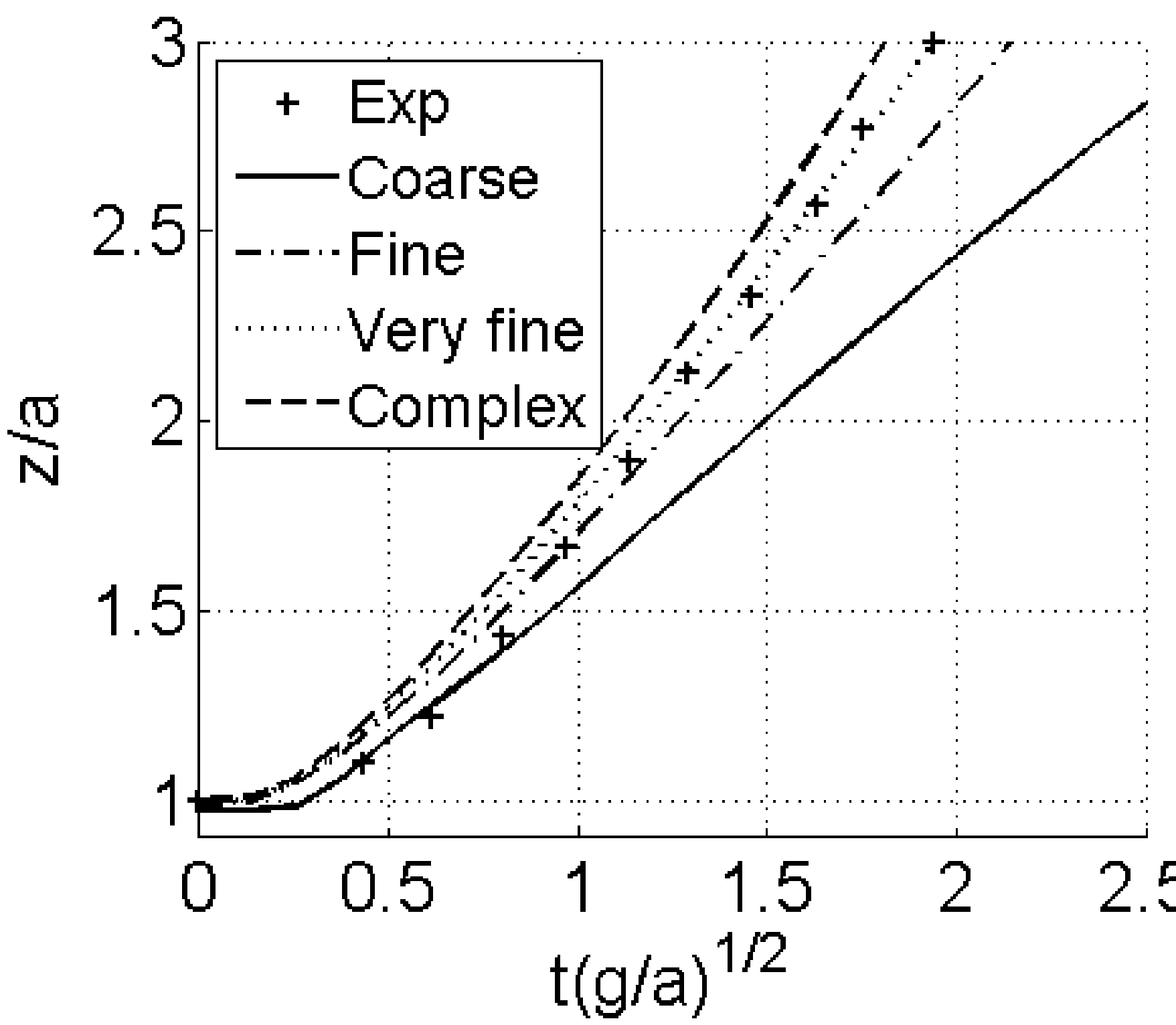
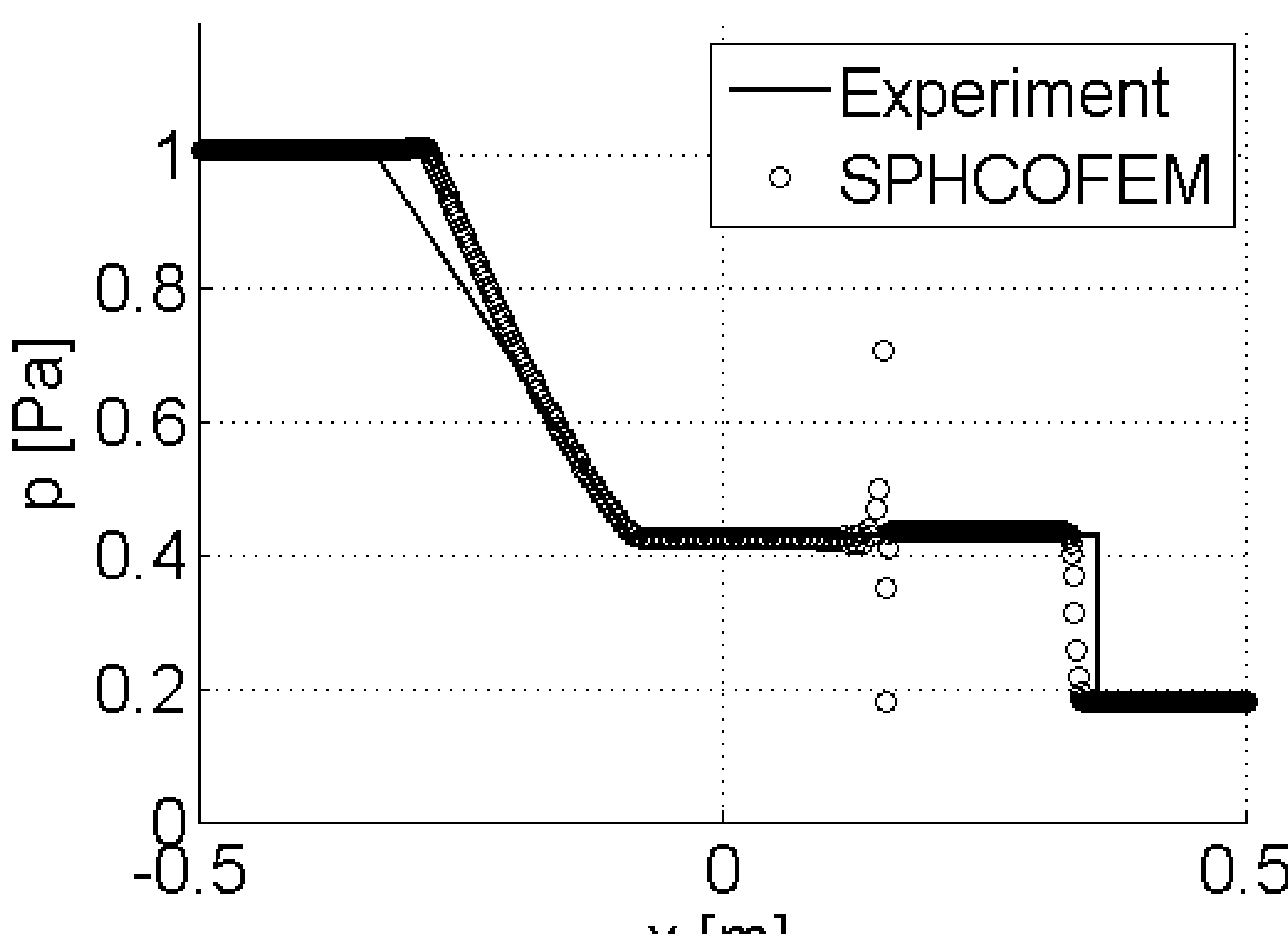
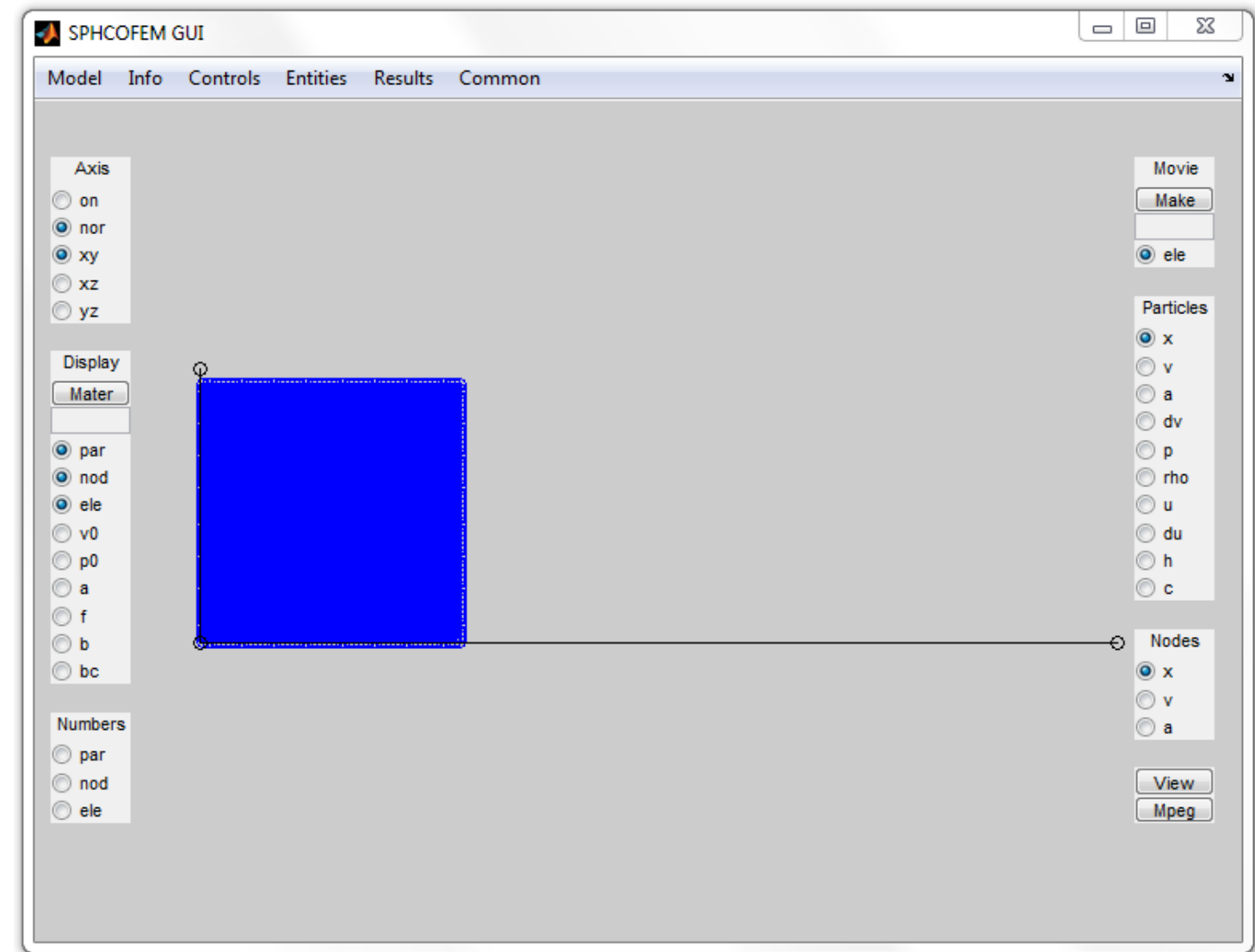
MATLAB is used for pre- and post-processing.

The presented algorithm is implemented as in-house software for fluid flow in 1D, 2D or 3D.

GUI is developed for reading the input file in the text format defined by keywords. The study shows both advantages and disadvantages using SPH coupled to FE and its exploitation for real engineering problems.

Acknowledgments

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Model	Particles	Variant	NNS	Time [s]
Coarse	19 x 19	SPH	N ²	1.34
		SPH	Subs	1.32
		X-SPH	N ²	5.40
		X-SPH	Subs	5.33
Fine	57 x 57	SPH	N ²	80.65
		SPH	Subs	56.00
		X-SPH	N ²	779.04
		X-SPH	Subs	754.14
Very fine	114 x 114	SPH	N ²	1 850.75
		SPH	Subs	974.21
Complex	228 x 228	SPH	N ²	79 243.73
		SPH	Subs	14 290.08